

24th June 2026

Flow of Incompressible Fluid
in Porous Media

$$q_B = -kA \frac{dp}{dl} \text{ (Darcy's unit)}$$

$$q_B = -0.001127 kA \frac{dp}{dl} \text{ (Field Unit)}$$

Field Units

$$k = \text{md}$$

$$p = \text{psi}$$

$$A = \text{ft}^2$$

$$L = \text{ft}$$

$$A = \text{ft}^2$$

$$B = \text{ab/STB}$$

$$q_B = -\frac{kA}{\mu} \frac{dp}{dl}$$

Types of flow in Porous Media

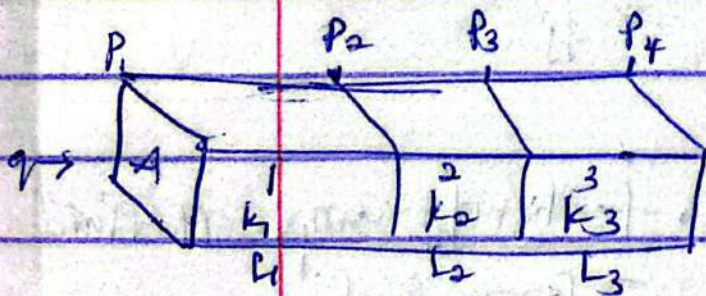
1. Incompressible

2. Slightly compressible (oil that contains some gas)

3. Compressible

1. Series Flow of Incompressible

Fluid in Porous Media



$$dp = q_B \frac{dl}{kA}$$

$$= 0.001127 kA$$

$$P_1 - P_4 = (P_1 - P_2) + (P_2 - P_3) + (P_3 - P_4)$$

$$\frac{q_B \mu L}{0.001127 k_A} = \frac{q_B \mu L_1}{0.001127 k_1 A} + \frac{q_B \mu L_2}{0.001127 k_2 A}$$

$$+ \frac{q_B \mu L_3}{0.001127 k_3 A}$$

$$0.001127 k_A$$

$$\left(\frac{q_B \mu}{0.001127} \right) \frac{L}{k_A} = \left(\frac{q_B \mu}{0.001127} \right) \left(\frac{L_1}{k_1} + \frac{L_2}{k_2} + \frac{L_3}{k_3} \right)$$

$$\left(\frac{L_1}{k_1} + \frac{L_2}{k_2} + \frac{L_3}{k_3} \right)$$

$$\frac{L}{k_{eq}} = \frac{L_1}{k_1} + \frac{L_2}{k_2} + \frac{L_3}{k_3} + \dots$$

$$k_{eq} = \frac{L}{\frac{L_1}{k_1} + \frac{L_2}{k_2} + \frac{L_3}{k_3} + \dots}$$

$$\frac{L_1}{k_1} + \frac{L_2}{k_2} + \frac{L_3}{k_3}$$

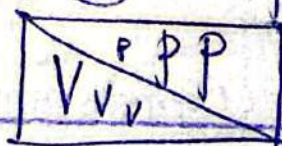
$$L = L_1 + L_2 + L_3 + \dots$$

$$L = \sum_{i=1}^n L_i$$

$$k_{eq} = \frac{\sum_{i=1}^n L_i}{\sum_{i=1}^n \frac{L_i}{k_i}}$$

2. Parallel Flow of Incompressible fluid
in Porous Media

Wednesday, 1st July 2026
Slightly Incompressible Fluid



$$C = -\frac{1}{V} \frac{dV}{dP}$$

$$-C dP = \frac{1}{V} dV$$

$$-C \int_{P_1}^{P_2} dP = \int_{V_1}^{V_2} \frac{1}{V} dV$$

$$-C (P_2 - P_1) = \ln V_2 - \ln V_1$$

$$C (P_1 - P_2) = \ln \left(\frac{V_2}{V_1} \right)$$

$$\frac{V_2}{V_1} = e^{C(P_1 - P_2)}$$

$$V_2 = V_1 e^{C(P_1 - P_2)} \quad \text{--- (1)}$$

C = compressibility, P_1^{-1}

$$P_1 = P_{\text{in}}$$

$$P_2 = P_{\text{out}}$$

$$J_2 = J_1 e^{C(P_1 - P_2)}$$

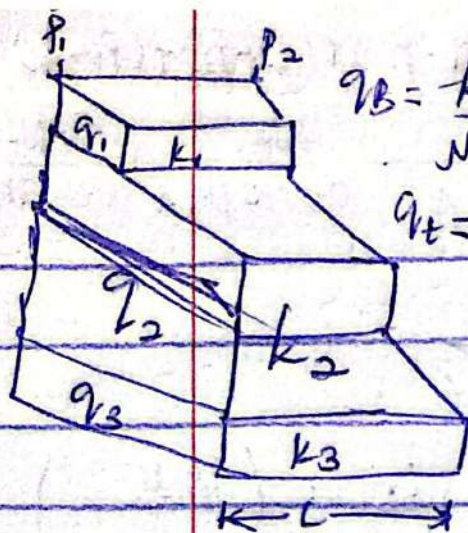
$$C = \frac{1}{e} \frac{dP}{dJ}$$

Flow of slightly compressible fluid
in Porous Media

Horizontal flow: $q_B = -0.001127 \frac{kA}{\mu} \left(\frac{dP}{dL} \right)$



(1)



$$q_B = \frac{kA}{\mu} \frac{dP}{dL}$$

$$q_t = q_1 + q_2 + q_3$$

Length is equal, μ & B are the same

$$q = 0.001127 \frac{kA}{\mu} \frac{dP}{dL}$$

$$B \mu dL$$

$$-0.001127 \frac{k_1 A_1}{\mu} dP$$

$$B \mu dL$$

$$= -0.001127 \frac{k_1 A_1}{\mu} dP - 0.001127 \frac{k_2 A_2}{\mu} dP$$

$$B \mu dL$$

$$B \mu dL$$

$$-0.001127 \frac{k_3 A_3}{\mu} dP$$

$$B \mu dL$$

$$= -0.001127 \frac{dP}{dL} (k_1 A_1 + k_2 A_2 + k_3 A_3)$$

$$B \mu dL$$

$$B \mu dL$$

$$(k_1 A_1 + k_2 A_2 + k_3 A_3)$$

$$k_{\text{eq}} A_t = k_1 A_1 + k_2 A_2 + k_3 A_3$$

$$k_{\text{eq}} = k_1 \frac{A_1}{A_t} + k_2 \frac{A_2}{A_t} + k_3 \frac{A_3}{A_t}$$

$$A_{\text{total}}$$

Inclined flow.

$$q_B = -0.001127 \frac{kA}{N} \left[\frac{dp}{dL} - 0.433 \gamma \cos \alpha \right]$$

$$velocity = \frac{V}{k}$$



From equation (1),

$$V = -0.001127 \frac{k}{\mu B} \left(\frac{dp}{dL} \right)$$

k = permeability, md (millidarcy)

A = Area, ft^2

L = ft

P = Pressure, psi

γ = Specific gravity of the fluid, no unit.

α = Angle of inclination

Continuation of eqn (2)

From Taylor Series,

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

If x is very small,

$$e^x \approx 1 + x$$

$$\Rightarrow e^{c(P_1 - P_2)} \approx 1 + c(P_1 - P_2)$$

$$q_B = V_1 [1 + c(P_1 - P_2)]$$

$$q_B = V_1 [1 + c(P_1 - P_2)]$$

In terms of density.

ρ = reservoir
 ρ_s = surface

Example:

1. A reservoir with an area of $47.8 ft^2$ and a length of 35 ft has a crude oil of viscosity $0.15 cP$ and formation volume factor of $1.025 b/b$ flowing through it with a differential pressure of 250 psi. If the permeability of the formation is 0.175 darcy, calculate:

a) the volumetric fluid flow through the reservoir

b) velocity of flow through the reservoir

$$A = 47.8 ft^2$$

$$L = 35 ft$$

$$\mu = 0.15 cP$$

$$B = 1.025 b/b$$

$$P = 250 psi$$

$$k = 0.175 d = 0.175 \times 10^{-3} md$$

$$q_B = -0.001127 \frac{kA}{N} \left(\frac{dp}{dL} \right)$$

$$= -0.001127 \times \frac{0.175 \times 10^{-3} \times 47.8 \times 250}{0.15}$$

$$= -448.9 = -448.9 \times 10^{-3} cfs$$

$$b) V = -0.001127 \frac{K}{\mu B} \left(\frac{dp}{dL} \right)$$

$$= -0.001127 \times 0.175 \times 10^{-3} \times 250$$

$$0.15 \times 1.025 \times 35$$

$$V = -9.1626 \times 10^{-5} \text{ ft}^3/\text{s}$$

$$FVF = \beta = \frac{\text{Volume in the reservoir}}{\text{Volume at the surface}}$$

$$= 10^{-8}$$

For slightly compressible fluid

$$q_B = q_R [1 + C(P_R - P)]$$

$$= -0.001127 \frac{KA}{\mu} \frac{dp}{dL}$$

(integration by part)

$$q_R dL = -0.001127 \frac{KA}{\mu} \frac{1}{1 + C(P_R - P)} dp$$

$$q_R \int_1^2 dL = -0.001127 \frac{KA}{\mu} \int_{P_1}^{P_2} \frac{1}{1 + C(P_R - P)} dp$$

$$q_R (L_2 - L_1) = -0.001127 \frac{KA}{\mu} \left[\ln(1 + C(P_R - P)) \right]_{P_1}^{P_2}$$

$$q_R (L_2 - L_1) = -0.001127 \frac{KA}{\mu} \left[\ln \left(\frac{1 + C(P_R - P_2)}{1 + C(P_R - P_1)} \right) \right]$$

$$q_R = \frac{-0.001127 KA}{L_2 - L_1} \ln \left[\frac{1 + C(P_R - P_2)}{1 + C(P_R - P_1)} \right]$$

P_R = Relative Pressure

Flow of compressible fluid in Porous Media

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \text{ Ideal gas}$$

$$\frac{P_1 V_1}{Z_1 T_1} = \frac{P_2 V_2}{Z_2 T_2} \text{ Real gas}$$

1. Reservoir condition, q_R

2. Surface condition, q_B

$$5.615 \text{ ft}^3 = 1 \text{ bbl}$$

$$B_g = \frac{V}{V_{sc}} = \frac{P_{sc} Z T}{P Z_{sc} T_{sc}} \text{ (ft}^3\text{)}$$

$$\frac{P V}{Z T} = \frac{P_{sc} V_{sc}}{Z_{sc} T_{sc}}$$

$$B_g = \frac{P_{sc} Z T}{5.615 P Z_{sc} T_{sc}}$$

$$P_{sc} = 14.7 \text{ psia}$$

$$T_{sc} = 520^\circ \text{R}$$

$$Z_{sc} = 1$$

$$B_g = 14.7 Z T$$

$$5.615 \times 1 \times 520 \times P$$

$$B_g = 0.005035 Z T$$

$$q_B = -0.001127 \frac{KA}{\mu} \frac{dp}{dx}$$

$$q (0.005035 Z T) = -0.001127 \frac{KA}{\mu} \frac{dp}{dx}$$

$$\frac{0.005035 - Z T q}{0.001127 P K A} = \frac{dp}{dx}$$

$$\frac{4.46762 \times 10^9}{kA} = -p \frac{dp}{dn}$$

$$\frac{4.46762 \times 10^9}{kA} \int_0^L dx = - \int_{P_1}^{P_2} p dp$$

$$\frac{4.46762 \times 10^9}{kA} L = - \left[\frac{p^2}{2} \right]_{P_1}^{P_2}$$

$$\frac{4.46762 \times 10^9}{kA} L = \frac{P_1^2 - P_2^2}{2}$$

$$q = 0.1119 kA \frac{(P_1^2 - P_2^2)}{2L}$$

in the numerator

$$\begin{cases} A = 75 \text{ ft}^2 & T = 85^\circ \text{F} \times \frac{5}{9} = 54.5^\circ \text{C} \\ L = 275 \text{ miles} \times \left(\frac{5280 \text{ ft}}{1 \text{ mile}} \right) = 1452000 \end{cases}$$

$$P_2 = 4500 \text{ psig} + 14.7 = 4514.7 \text{ psia}$$

$$P_1 = 275 \text{ psig} + 14.7 = 289.7 \text{ psia}$$

$$q = 0.1119 \times 820 \times 75 \times \left(\frac{289.7^2 - 4514.7^2}{2 \times 1452000} \right)$$

$$q = -10.1233 \text{ bbl/day}$$

$$v = \frac{q}{A} = \frac{-10.1233 \text{ bbl}}{\text{day}} \times \frac{1}{75 \text{ ft}^2} \times \frac{15.615 \text{ ft}^3}{1 \text{ bbl}}$$

$$= -0.135 \times 15.615$$

$$= -0.758 \text{ ft/day}$$

Example 2:

A natural gas with $U = 0.25 \text{ cp}$ and compressibility factor of 0.93 flows in a reservoir with an area of 75 ft^2 at a temperature of 85°F from 275 miles from the well.

If the pressure at the outlet of the well face is 4500 psig and the pressure at the inlet is 275 psig, calculate the

(a) volume of gas which the gas flows towards the well (b) velocity at which the gas flows towards the well.

$$U = 0.25 \text{ cp} \quad k = 0.82 \text{ d} = 0.82 \times 10^3 \text{ md} = 820 \text{ md}$$

Solution

$$U = 0.25 \text{ cp}, \quad Z = 0.93$$

Pressure above 2000 psi - compressible

~~NZ~~
(cp)

Gradient = $\frac{dZ}{p}$
 For pressure below
 2000 psi
 $dZ = \text{constant}$

$$0 \leq p \leq 2000 \text{ psi}$$
$$JZ = \text{constant}$$

$$V_B = -0.001127 \text{ kA} \cdot \text{dp}$$

1142

$$b = \frac{V}{V_{sc}} \Rightarrow \frac{P_V}{Z_T} = \frac{P_{so} V_{so}}{Z_{so} T_{so}}$$

$$B = \frac{V_{sc}}{I_{sc}} = \frac{P_{sc}}{I_{sc}^2 R_p}$$

$$\underline{Y} = 14.727 = 0.02827 \text{ TP}$$

V_{sc}	1×520	ϕ
$0.02827 \approx 1$	$q = -0.001127 \text{ KA}$	$d\phi$
ϕ		$N \frac{d\phi}{dt}$

$$\frac{0.02827}{p} \frac{dT}{T} = -0.001127 \frac{K}{dx} dx$$

$$\frac{0.62827 \text{ ZMT}}{\varphi} \int_0^L f_1 = 0.00127 \text{ kg} \frac{P}{\text{kg}} \frac{P}{\text{kg}}$$

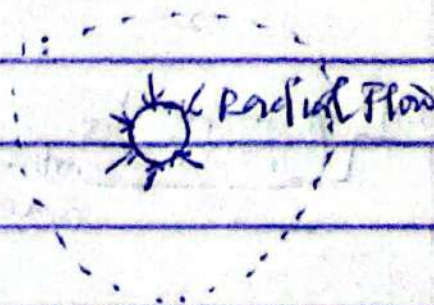
$$\frac{0.02827 \text{ Z}}{P} T_{q,L} = -0.0012714 \text{ (P}_2 - P_1)$$

0-02827-2 NTL

$q = -0.0399 \text{ PKA } (P_2 - P_1)$ for $P > 200$

EUTL

Rachal Flow of Incompressible Fluid
in a Linear system


$$r_W = \text{radius of } \omega$$

$r_o =$ radius of outside

P_e = pressure at the boundary

P_w = Pressure at the well

$$q_B = -0.00127 \text{ kA} \frac{dq}{dn}$$

$$\frac{q_B}{A} = -0.001127 \text{ kPa}$$

$$\frac{q_D}{2\pi \cdot h} = -0.00127 \frac{\text{K}}{\text{m}} \frac{dP}{dr}$$

$$\frac{q_B}{2\pi h} \int_{r_w}^{r_e} \frac{1}{r} dr = -0.00127 k \left(\frac{Pe}{\mu P_w} \right)$$

$$\frac{q_B \ln r_B}{h r_B} = -0.00127 \times 2\pi \times K (P_1 - P_2)$$

$$\frac{q_B}{h} \ln\left(\frac{r_e}{r_w}\right) = 0.00708 \frac{kh}{B\mu} (P_w - P_e)$$

$$\frac{q_B}{h} \ln\left(\frac{r_e}{r_w}\right) = \frac{q = 0.00708 kh (P_w - P_e)}{B\mu \ln\left(\frac{r_e}{r_w}\right)}$$

Radial Flow of slightly
compressible fluid

$$q_B = -0.001127 \frac{kA}{\mu} \frac{dp}{dr}$$

$$q_R = q_R (1 + c(P_R - p))$$

$$q_R (1 + c(P_R - p)) = -0.001127 \frac{kA}{\mu} \frac{dp}{dr}$$

$$\frac{q_R \mu}{k 2\pi h} \int_{r_w}^{r_e} \frac{1}{r} dr = -0.001127 \int_{P_w}^{P_e} \frac{dp}{1 + c(P_R - p)}$$

$$\frac{q_R \mu}{kh} \ln\left(\frac{r_e}{r_w}\right) = \frac{-0.001127 \times 2\pi}{c} \left[\frac{1 + c(P_R - P_2)}{1 + c(P_R - P_1)} \right]$$

$$\frac{q_R}{kh} \mu \ln\left(\frac{r_e}{r_w}\right) = \frac{0.001127}{c} \left[\frac{1 + c(P_R - P_2)}{1 + c(P_R - P_1)} \right]$$

1 Radial flow of compressible fluid
(steady-state)

$$\frac{q_{TSC}}{T_{sc}} Z \ln \left(\frac{r_e}{r_w} \right) = -0.00708 \frac{k h}{\mu} (P_e^2 - P_w^2)$$

$$q_B = - \frac{0.001127 k A}{\mu} \frac{dp}{dr}$$

$$q = \frac{-0.00254 k h (P_e^2 - P_w^2) T_{sc}}{P_{sc} \ln \left(\frac{r_e}{r_w} \right) Z P}$$

$$\frac{q}{2\pi r h \times P} = - \frac{0.001127 k}{\mu} \frac{dp}{dr}$$

$$q \int_r^{r_e} \frac{dr}{r} = - \frac{0.001127 k}{\mu} \int_{P_w}^{P_e} \frac{P}{P} dP$$

$$q \ln \left(\frac{r_e}{r_w} \right) = - \frac{0.00708 k h}{\mu} (P_e - P_w)$$

$$q = \frac{-0.00708 k h T_{sc}}{\mu \ln \left(\frac{r_e}{r_w} \right) P_{sc}} (P_e - P_w)$$

$$q_B = - \frac{0.001127 k A}{\mu} \frac{dp}{dr}$$

$$B = V = \frac{P_{sc} Z T_{sc}}{P}$$

$$\frac{V}{Z} B = \frac{P_{sc}}{T_{sc}} \left(\frac{T}{P} \right)$$

$$\frac{q P_{sc}}{T_{sc}} \left(\frac{Z T}{P} \right) = - \frac{0.001127 k A}{\mu} \frac{dp}{dr}$$

$$\frac{0.9 P_{sc}}{2\pi h T_{sc}} \int_{r_w}^{r_e} \frac{dr}{r} = - \frac{0.001127 k}{\mu} \int_{P_w}^{P_e} \frac{dP}{P}$$

22nd July 2022

Radial Flow of Compressible Fluid (Steady state)

$$q_r = -0.001127 kA \frac{dp}{dr}$$

$$\frac{q_r P_{sc} Z T}{5.615 \times 10^4} = 0.001127 kA \frac{dp}{dr}$$

p is less than 2000, $\mu Z = \text{constant}$

$$\frac{q_r P_{sc} \mu Z T}{T_{sc} k h} \int_{r_w}^{r_e} \frac{1}{r} dr = \int_{P_w}^{P_e} p dp$$

$$= -0.001127 \times 5.615 \times 2\pi \int_{P_w}^{P_e} p dp$$

$$\frac{q_r P_{sc} \mu Z T}{T_{sc} T_{sc} k h} \ln \frac{r_e}{r_w} = 0.39766 \left(\frac{P_e^2 - P_w^2}{2} \right)$$

k_{eq} = capacity of bed / station

$$q_T = q_1 + q_2 + q_3$$

$$h_T = h_1 + h_2 + h_3$$

$$\frac{0.00708 k_{eq} h_T (P_w - P_e)}{\mu Z \ln \left(\frac{r_e}{r_w} \right)}$$

$$q = \frac{0.00708 k_{eq} h_T (P_w - P_e)}{\mu Z \ln \left(\frac{r_e}{r_w} \right)}$$

$$\left[\frac{0.00708 k_{eq} h_T (P_w - P_e)}{\mu Z \ln \left(\frac{r_e}{r_w} \right)} \right]$$

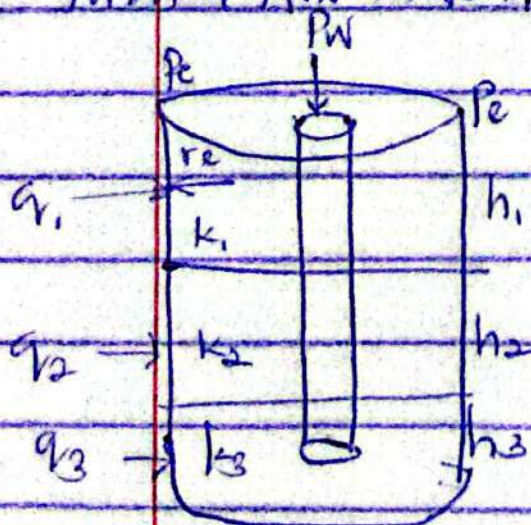
$$k_{eq} h_T = k_1 h_1 + k_2 h_2 + k_3 h_3$$

$$k_{eq} = \frac{k_1 h_1 + k_2 h_2 + k_3 h_3}{h_T = h_1 + h_2 + h_3}$$

$$k_{eq} = \frac{\sum k_i h_i}{\sum h_i}$$

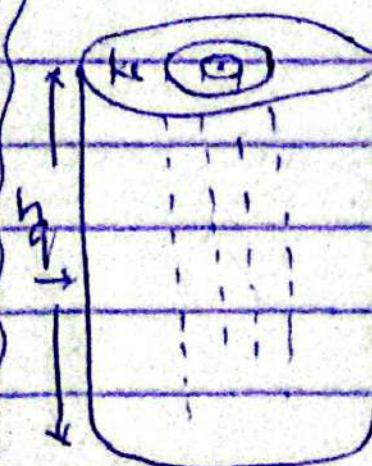
Calculation of Average Permeability

of Parallel Flow in Radial Direction



$$q_2 = \frac{0.00708 k_2 h_2 (P_w - P_e)}{\mu Z \ln \left(\frac{r_e}{r_w} \right)}$$

Calculation of Average Permeability of Series Flow in Radial Direction



$$q = \frac{0.00708 kh (P_w - P_e)}{\mu z \ln\left(\frac{r_e}{r_w}\right)}$$

$$P_w - P_e = \frac{q \mu z \ln\left(\frac{r_e}{r_w}\right)}{0.00708 kh}$$

$$P_w - P_{e3} = (P_w - P_{e1}) + (P_{e1} - P_{e2}) + (P_{e2} - P_{e3})$$

$$\frac{q \mu z \ln\left(\frac{r_{e3}}{r_w}\right)}{0.00708 kh} = \frac{\ln\left(\frac{r_e}{r_w}\right)}{k_1} + \frac{\ln\left(\frac{r_{e2}}{r_{e1}}\right)}{k_2} + \frac{\ln\left(\frac{r_{e3}}{r_{e2}}\right)}{k_3}$$